

RESEARCH ARTICLE

From Static to Dynamic: Evaluating the Perceptual Impact of Dynamic Elements in Urban Scenes Using Generative Inpainting

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ABSTRACT

Understanding urban perception from street-view imagery has become a central topic in urban analytics and human-centered urban design. However, most existing studies treat urban scenes as static and largely ignore the role of dynamic elements such as pedestrians and vehicles, raising concerns about potential bias in perception-based urban analysis. To address this issue, we propose a controlled framework that isolates the perceptual effects of dynamic elements by constructing paired street-view images with and without pedestrians and vehicles using semantic segmentation and generative inpainting powered by large vision models. Based on 720 paired images from Dongguan, China, a perception experiment was conducted in which participants evaluated original and edited scenes (removal of pedestrians and vehicles) across six perceptual dimensions, including *wealth*, *safety*, *vibrancy*, *beauty*, *boredom*, and *depression*. The results indicate that removing dynamic elements leads to a consistent decrease in perceived vibrancy (30.97%), whereas changes in other dimensions are more moderate and heterogeneous. To further explore the underlying mechanisms, we trained 11 machine-learning models using multimodal visual features. Feature-importance analysis identifies lighting conditions, human presence, and depth variation as key factors driving perceptual change. At the individual level, 65% of participants exhibited significant vibrancy changes, compared with 35%–50% for other dimensions, revealing notable inter-individual sensitivity; gender further showed a marginal moderating effect on safety perception. Beyond controlled experiments, the trained model was extended to a city-scale dataset covering 47,963 locations (191,852 images) to predict vibrancy changes after the removal of dynamic elements. The city-level results reveal that such perceptual changes are widespread and spatially structured, affecting 73.7% of locations and 32.1% of images, suggesting that urban perception assessments based solely on static imagery may substantially underestimate urban liveliness. Overall, this study highlights the critical role of dynamic elements in shaping urban perception and underscores the importance of accounting for transient urban features in large-scale perception-driven urban studies.

1 | Introduction

Urban perception plays a central role in shaping how individuals experience and emotionally engage with their surroundings

(Ito et al. 2024; Zhu et al. 2025). As cities become increasingly dense, diverse, and visually complex, understanding how people interpret and respond to urban environments has become a key concern for urban planning, landscape architecture,

and public health (Cao et al. 2025). The way people perceive a city affects their emotional states, such as feeling safe, and also influences long-term behaviors, place attachment, and well-being (Wolch et al. 2014). Visual stimuli in the urban environment, including buildings, greenery, signage, and people, play a crucial role in shaping these perceptions (Greene and Oliva 2009; Lindal and Hartig 2013). Recent advances in computing and spatial analysis have made it possible to study these perceptions at scale, revealing how urban form and visual features correlate with feelings of wealth, safety, vibrancy, beauty, boredom, and depression (Dubey et al. 2016; Zhang et al. 2018). These developments have supported a growing body of data-driven research on urban perception, which not only enables empirical analysis of subjective spatial experience but also provides critical tools for improving the livability and sustainability of cities (Wu et al. 2022).

Building on this foundation, researchers have increasingly adopted methods based on street view imagery, semantic segmentation, and machine learning to quantify how specific urban elements shape emotional responses. For instance, studies have used deep learning to extract visual features from images and train models to predict human judgments across multiple perceptual dimensions (Dubey et al. 2016; Zhao et al. 2023; Liang et al. 2024). Others have employed large-scale crowdsourced evaluations, such as the Place Pulse project (Salesses et al. 2013), a global collection of street images in 688 cities (Hou et al. 2024), or proposed adversarial learning frameworks to economically assess urban perceptions in Chinese cities (Yao et al. 2019). Some approaches also incorporate spatial syntax or visual hierarchy to enhance the interpretability of perception scores (Wang et al. 2022). More recently, researchers have integrated eye-tracking data to provide a more nuanced understanding of how people visually explore city scenes and which elements attract attention (Noland et al. 2017; Liao et al. 2019; Wu et al. 2022; Yang, Deng, Hu, Chao, et al. 2024; Kang et al. 2026). These methods capture visual saliency and attentional focus, enabling a richer model of how urban features such as cars, sidewalks, trees, or crowds influence perception.

Despite these advances, a key limitation persists: most existing studies rely on static street view images, which represent only a single moment in time and fail to capture the inherent temporal dynamics of urban environments (Yang, Deng, Hu, Guan, et al. 2024). In real-world settings, transient elements such as pedestrians, vehicles, and street activities constantly reshape how people experience the city (Vater et al. 2022; Qin et al. 2025). These moving objects not only contribute to the semantic meaning of a scene but also dominate the visual field and emotional interpretation. However, their role has largely been overlooked in existing computational models, which tend to treat the visual environment as static and uniform (Yao et al. 2025). A fundamental question, therefore, arises: how can we generate or manipulate dynamic urban scenes in a controlled way to better understand their perceptual effects? The recent progress of generative models provides a promising opportunity. Models such as Stable Diffusion (Rombach et al. 2022), DALL-E 3 (Betker et al. 2023), and LaMa (Suvorov et al. 2022) can synthesize high-quality urban scenes that include or exclude specific dynamic elements, such as

pedestrians or vehicles. By leveraging these generative capabilities, researchers may create large-scale, semantically consistent datasets to test how dynamic features influence human perception of cities.

Based on the above considerations, this study explores how the presence or absence of dynamic elements affects people's perception of urban street scenes. In this study, "dynamic elements" specifically refer to transient objects, namely pedestrians and vehicles, that commonly appear within a single street-view scene, as opposed to temporal variations across different times of day or seasons. Our focus is on how the presence or absence of these moving objects within a static image influences perceptual judgments, rather than on changes in the scene over time. Regarding urban perception, we follow established frameworks in urban studies and environmental psychology and characterize it along two major psychological axes: affective valence (positive–negative emotional response) and perceived urban activity or intensity (Salesses et al. 2013; Dubey et al. 2016; Wu et al. 2022). Building on this perspective, six perceptual dimensions are selected to jointly span these axes. Specifically, *wealth*, *safety*, and *beauty* capture stable evaluative judgments of urban environments, while *vibrancy* reflects perceived urban activity and social presence. In contrast, *boredom* and *depression* represent negative affective responses associated with low stimulation or adverse urban conditions. We employ StabilityAI's inpainting tool to remove dynamic elements from real street-view images while preserving spatial and semantic coherence and construct a paired dataset of 720 original and edited street scenes. Forty participants were then invited to rate urban perception across these image pairs under randomized conditions, without being informed that each pair consisted of the same scene under different conditions, thereby ensuring that they were unaware of the pairing design. We analyze these results at both the image and individual levels to identify consistent perceptual differences induced by the presence or absence of dynamic elements. Machine learning models are further applied to explore which visual characteristics of street scenes contribute most to these differences and to predict perceptual variations at the city scale.

2 | Related Works

Urban perception refers to the emotional experiences and cognitive evaluations individuals form in response to urban environments, encompassing feelings such as safety, vibrancy, beauty, and wealth (Dubey et al. 2016). Understanding these subjective experiences has become increasingly important for informing people-centered urban design, promoting mental health, and enhancing urban livability. Since studying urban perception through street imagery requires systematically acquiring data, applying analytical methods, and revealing the cognitive mechanisms underlying human responses, existing research has primarily developed along these three dimensions: (1) *data sources*, (2) *mechanism exploration*, and (3) *analytical approaches*. Below, we review each dimension in an order that mirrors the research workflow of the present study—from data collection, through perceptual measurement, to computational modeling—while progressively building the case that dynamic elements have been largely overlooked.

2.1 | Data Source

In terms of data, early studies relied heavily on traditional approaches such as surveys and interviews to capture individuals' perceptions of urban spaces (Kaplan and Kaplan 1989; Nasar 1990). Although valuable, these methods were often constrained by limited geographic coverage and subjective biases. The introduction of large-scale street view imagery, such as Google Street View and Baidu Street View, marked a major shift, enabling researchers to systematically collect visual representations of urban environments across diverse settings (Wang et al. 2025). Landmark projects such as Place Pulse (Salesses et al. 2013) compiled global datasets through crowdsourced pairwise comparisons, allowing researchers to link perceptions of safety, wealth, vibrancy, beauty, and activity to specific visual stimuli. Further expansions through Place Pulse 2.0 (Dubey et al. 2016) broadened the geographic scope and perceptual dimensions, providing a robust foundation for computational modeling. Other datasets, such as the UrbanPerception dataset, the Global Streetscapes dataset (Hou et al. 2024), and a time-series street view imagery dataset for analyzing urban physical disorder in China (Ma et al. 2025), have further enriched the available resources for studying the visual and emotional perception of cities. In addition, Wang et al. (2025) evaluated the data quality and temporal availability of street view imagery from Google and Baidu. Critically, however, all these data sources capture urban scenes as static snapshots, providing no information about transient elements such as pedestrians and vehicles that may substantially influence how people perceive a scene. This limitation motivates our use of street view imagery as a base data source, while our methodological contribution lies in systematically manipulating these images to isolate the perceptual effects of dynamic elements.

2.2 | Mechanism Exploration

Before turning to computational models, it is essential to understand what is known about the cognitive mechanisms through which urban visual features shape perception. Initial studies inferred attention and emotional responses indirectly through aggregated user ratings (Salesses et al. 2013; Yao et al. 2019). However, advances in physiological sensing technologies have enabled the direct capture of cognitive indicators such as eye movements (Noland et al. 2017; Liao et al. 2019; Wu et al. 2022; Yang, Deng, Hu, Chao, et al. 2024; Yang, Deng, Hu, Guan, et al. 2024) and, more recently, electroencephalography (EEG) signals (Hei et al. 2025; Yang et al. 2025). Eye-tracking experiments have demonstrated that attentional allocation to specific urban elements—such as pedestrians, vehicles, and vegetation—significantly influences perceived safety, vibrancy, and social presence. Complementarily, EEG studies have begun to reveal how different urban visual stimuli evoke distinct neural responses associated with emotional and cognitive processing. Relatedly, Navarrete-Hernandez and Zegras (2023) found that travelers' affective subjective well-being is primarily shaped by experiential details such as environmental comfort, perceived safety, and waiting reliability, rather than by travel efficiency alone, highlighting that human-centered design must attend to the full user experience. While these physiological approaches offer valuable insights into automatic and pre-conscious

cognitive processes, our study focuses on explicit, conscious perceptual judgments that are most directly relevant to urban design and planning decisions. Accordingly, we adopt Likert-scale survey evaluations to capture participants' deliberate assessments across a large set of stimuli (720 images $\times$ 6 dimensions), enabling systematic comparison between original and edited scenes at scale. This resonates with our argument that urban perception depends not only on static built structures but also on dynamic, human-centered visual elements that serve as essential experiential cues. These multimodal insights mark an important step toward a deeper, cognitively grounded understanding of how urban environments are perceived.

2.3 | Analytical Approaches

Building on these rich data resources, the analytical methods for studying urban perception have evolved substantially. Early computational efforts mainly employed traditional machine learning techniques, such as support vector machines (Naik et al. 2014) and random forests (Quercia et al. 2014), often relying on handcrafted visual features extracted from images. *These models take manually calculated image attributes (color proportions, building and vegetation area ratios, depth statistics) as inputs and output predicted human perception scores for urban dimensions like safety and vibrancy.* With the rise of deep learning, convolutional neural networks (CNNs) and related architectures were increasingly applied to directly learn perceptual patterns from raw visual inputs (Dubey et al. 2016; Zhang et al. 2018; Fan et al. 2023; Ye et al. 2024). *CNNs use unprocessed street-view image pixels as input and directly output regression values matching people's subjective perceptual ratings without manual feature extraction.* More recently, adversarial learning frameworks (Yao et al. 2019) and spatial syntax-based models (Wang et al. 2022) have been introduced to enhance prediction robustness and interpretability, reflecting a broader trend toward more sophisticated, end-to-end perceptual modeling. *Adversarial learning models take paired street images and crowdsourced comparison labels as input to output relative perceptual rankings between scenes; spatial syntax models combine street image visual features and road network spatial indicators as dual inputs to generate explainable perception scores.* Alongside these developments, the emergence of foundation models has further broadened the methodological landscape, with large vision models increasingly being applied to tasks such as urban functional zone classification and poverty inference (Huang et al. 2024; Wu et al. 2025). *Large vision models use static street-view images paired with text prompts as input and output zero-shot predictions of urban functional attributes and perceptual tendencies.*

However, despite this methodological diversity, existing models are trained and evaluated on static street-view images, meaning they implicitly assume that perceptual judgments derived from these images are representative of how people experience real urban environments. This assumption has not been systematically tested. *To address this unvalidated static-scene bias, our research requires a classifier that can compare paired original and edited street images to quantify perceptual shifts caused by removing dynamic elements and trace their visual drivers. We therefore seek an algorithm with balanced prediction performance, high computing efficiency, and reliable interpretability, which is the*

Stimulus preparation Perceptual dimension and participants

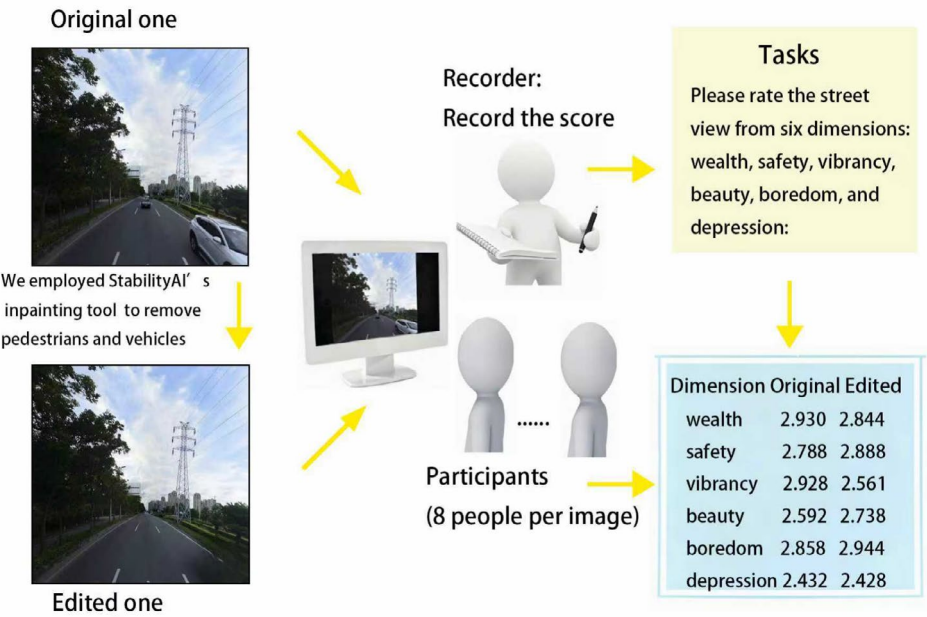


FIGURE 1 | Experimental procedure.

core reason we adopt XGBoost in this work. In our study, we select XGBoost as our primary classifier because it offers an optimal balance of predictive accuracy, computational efficiency, and interpretability—the latter being essential for identifying which visual features (e.g., lighting, depth, semantic categories) drive perceptual changes. The multimodal feature set we employ (color, depth, and semantic features) is specifically designed to capture the structural and visual changes introduced by the removal of dynamic elements, allowing us to move beyond prediction toward mechanistic explanation.

In summary, although substantial progress has been made in data collection, cognitive measurement, and computational modeling of urban perception, a critical gap remains: existing studies treat urban scenes as static and fail to account for the perceptual influence of dynamic elements such as pedestrians and vehicles. The data sources provide only static snapshots; the mechanistic evidence shows that human presence matters but has not been systematically manipulated; and the analytical models are trained on static images without testing their sensitivity to the presence or absence of transient objects. The present study directly addresses this gap by (1) using generative inpainting to construct paired static versus dynamic street-view images, (2) conducting controlled perception experiments with Likert-scale ratings across six dimensions, and (3) applying XGBoost with multimodal features to identify the visual drivers of perceptual change—thereby offering the first systematic investigation of how dynamic elements shape urban perception.

3 | Methodology

Figure 1 presents the experimental procedure of this study, which consists of two main processes. The first process focuses

on stimulus preparation, in which large-scale street-view images are collected and paired stimuli are constructed by selectively removing dynamic elements, including pedestrians and vehicles, through semantic segmentation and generative inpainting (Section 3.1). The second process focuses on collecting perceptual evaluations from participants on both the original and edited scenes across multiple perceptual dimensions (Section 3.2). Together, these two processes form the basis for subsequent image-level and individual-level analyses.

3.1 | Stimulus Preparation

To construct the experimental stimuli, we first extracted the road network of Dongguan, China, from OpenStreetMap (OSM). The location of Dongguan is shown in Figure 2. Dongguan is almost the fastest urbanizing city in southern China, which has received limited attention in street-view-based urban perception studies and was therefore selected as the study area. At 50-m intervals along the road network, panoramic street-view images were retrieved from Baidu Maps, yielding 47,963 locations collected between February 10 and February 14, 2025. To minimize distortions and artifacts introduced by the street-view capture vehicle, the camera perspective was adjusted by tilting the view upward by 15 degrees. From each panoramic image, four directional views (front, back, left, and right) were extracted to construct the initial street-view dataset, resulting in a total of 191,852 images. Subsequently, semantic segmentation was performed using DINOv2 (Oquab et al. 2023), a powerful foundation model released by Meta AI. The segmented visual elements include sky, trees, grass, buildings, roads, sidewalks, overpasses, vehicles, pedestrians, railings, walls, streetlights, and signposts. This categorization follows established practices in computational urban perception studies, which have demonstrated that these visual components play distinct roles

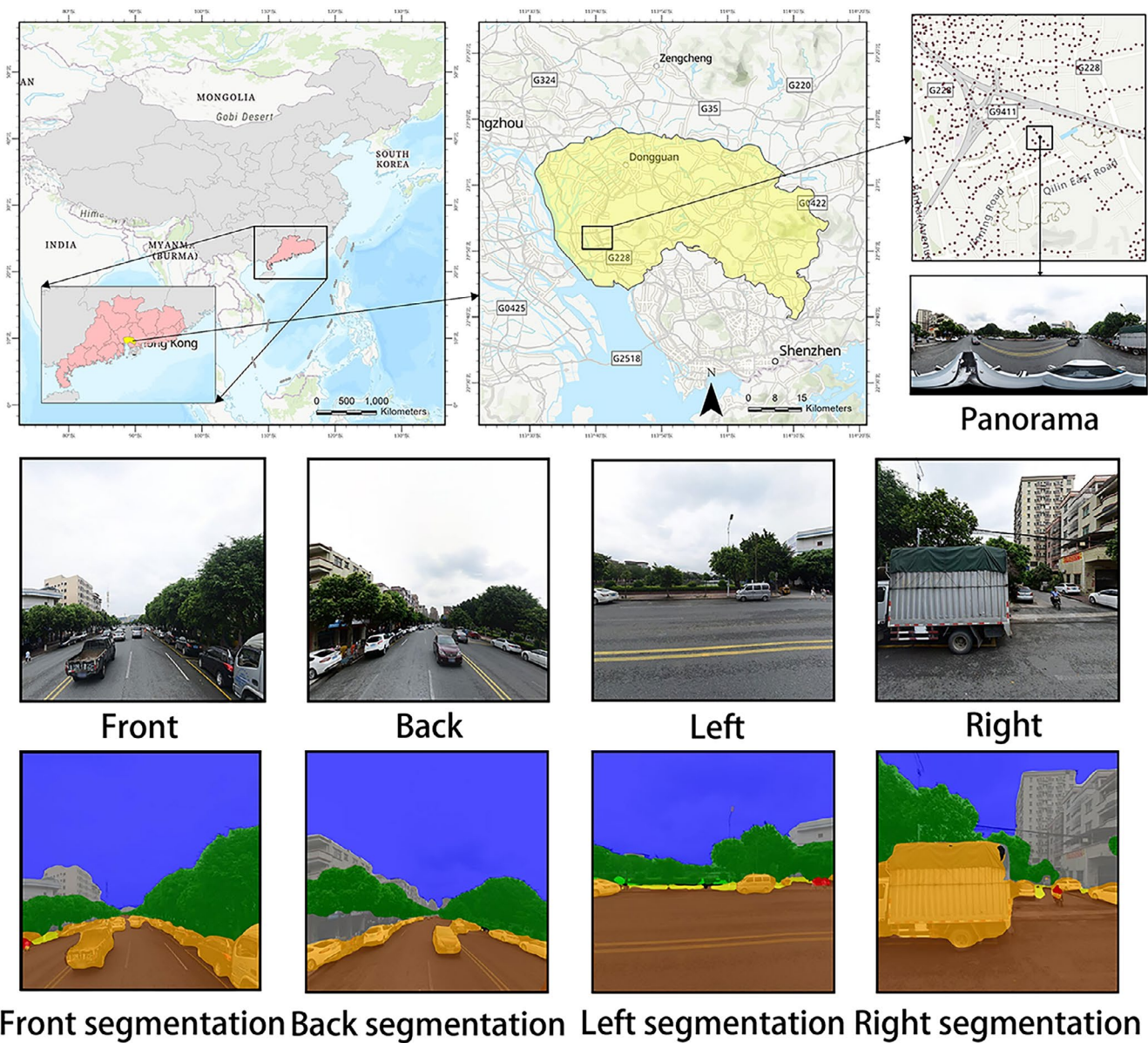


FIGURE 2 | Data source and image segmentation.

in shaping human judgments of urban environments (Yang, Deng, Hu, Guan, et al. 2024; Cao et al. 2025). The overall distribution of the segmented visual elements is summarized in Table 1.

As shown in the table, we can observe that static elements such as sky (mean proportion: 32.53%), roads (21.57%), buildings (15.02%), and trees (16.72%) dominate the visual composition of urban scenes, reflecting the structural characteristics of the built environment. Notably, dynamic elements are also highly prevalent: vehicles appear in 78.85% of images, while pedestrians are present in 25.78% of images. These results indicate that dynamic elements such as vehicles and pedestrians also constitute a non-negligible component of street-view imagery and therefore warrant systematic investigation.

To systematically construct paired stimuli with and without dynamic elements, we then design a three-stage pipeline for dynamic element removal and quality control.

3.1.1 | Step 1: Dynamic Element Detection and Mask Generation

We first apply DINOv2-based semantic segmentation to each street-view image to identify dynamic elements (Oquab et al. 2023), specifically pedestrians and vehicles. The segmentation results are used to generate binary masks indicating the spatial locations of these dynamic objects. Images in which dynamic elements account for more than 5% of the visual area are retained as candidate scenes, resulting in a dataset enriched with dynamic content.

3.1.2 | Step 2: Mask-Guided Diffusion-Based Inpainting

For each candidate image, the generated binary masks are combined with the original image and fed into StabilityAI's Stable Diffusion-based inpainting API. The inpainting process is conditioned on (a) the original street-view image, (b) the

TABLE 1 | Results of the visual element segment.

Element	Frequency	Maximum	Minimum	Mean
Sky	95.56%	77.32%	0.00%	32.53%
Trees	93.07%	99.99%	0.00%	16.72%
Grass	49.10%	66.63%	0.00%	1.24%
Buildings	89.97%	100.00%	0.00%	15.02%
Roads	97.17%	53.45%	0.00%	21.57%
Sidewalks	82.85%	53.99%	0.00%	3.25%
Overpass	6.64%	70.02%	0.00%	0.23%
Vehicles	78.85%	98.29%	0.00%	3.51%
Persons	25.78%	34.25%	0.00%	0.08%
Railings	8.72%	33.62%	0.00%	0.07%
Walls	53.68%	100.00%	0.00%	1.43%
Streetlights	37.02%	3.32%	0.00%	0.03%
Signposts	44.69%	38.70%	0.00%	0.25%

segmentation mask, and (c) a textual prompt (“remove pedestrians and vehicles from the image”). This step generates edited images in which dynamic elements are removed while preserving global spatial structure and semantic consistency.

3.1.3 | Step 3: Human Validation and Quality Filtering

To ensure dataset reliability, we conduct a manual quality inspection of all generated images. The validation is performed by three graduate students majoring in Geographic Information Science (GIS), who independently review each image. Only images that achieve full agreement among all three annotators are retained for further analysis. We identify three main categories of failure cases: (a) incomplete detection, where certain dynamic objects (e.g., bicycles or partially occluded pedestrians) are not fully removed due to segmentation misses; (b) large blurred regions or structural artifacts, often appearing at image boundaries or in regions with complex geometry; and (c) visible removal traces, including inconsistent blending, texture distortion, or lighting artifacts that affect scene realism. Images exhibiting any of these issues are excluded from further analysis.

In total, 968 images are generated through the inpainting pipeline, among which 720 images pass the quality filtering stage and are retained for subsequent experiments. The excluded images consist of 57 cases of incomplete detection, 89 cases of large blurred regions, and 102 cases of visible artifact traces. The filtering process is terminated once a sufficient number of high-quality paired samples is obtained. Representative examples of valid outputs and the three types of failure cases are shown in Figure 3. The final dataset is drawn from different view directions, including front (201 images), back (214 images), left (154 images), and right (151 images), ensuring a relatively balanced spatial distribution. For experimental use, the dataset is further organized into 10 groups of original images (72 images each)

and 10 corresponding groups of edited images, where each pair represents the same scene with and without dynamic elements.

3.2 | Perceptual Dimensions and Participants

3.2.1 | Perceptual Dimensions

Following common practice in urban perception research and grounded in environmental psychology theory (e.g., Saleses et al. 2013; Dubey et al. 2016; Wu et al. 2022), participants were asked to evaluate each scene along six perceptual dimensions: wealth, safety, vibrancy, beauty, boredom, and depression. These dimensions can be interpreted within a broader framework of affective valence and perceived urban vitality. Specifically, wealth reflects the perceived socioeconomic status of urban environments; safety captures perceived risk and environmental order, consistent with environmental criminology and Crime Prevention through Environmental Design theory (Crowe and Fennelly 2013); beauty reflects aesthetic preference and visual harmony. Vibrancy is associated with perceived urban vitality and human activity intensity, conceptually linked to “urban life” in classic urban theory (e.g., Gehl 2012). Boredom and depression represent negative affective states corresponding to low arousal and negative valence in dimensional emotion models. These dimensions have been widely adopted to capture both positive and negative aspects of emotional and cognitive responses to urban environments, as also demonstrated in Place Pulse 2.0 (Dubey et al. 2016). For each image, participants provided ratings on a five-point Likert scale (1 = very low, 5 = very high) across all six dimensions.

3.2.2 | Participants

A total of 40 participants (12 male, 28 female; aged from 20 to 32; all with normal or corrected-to-normal vision and no ocular

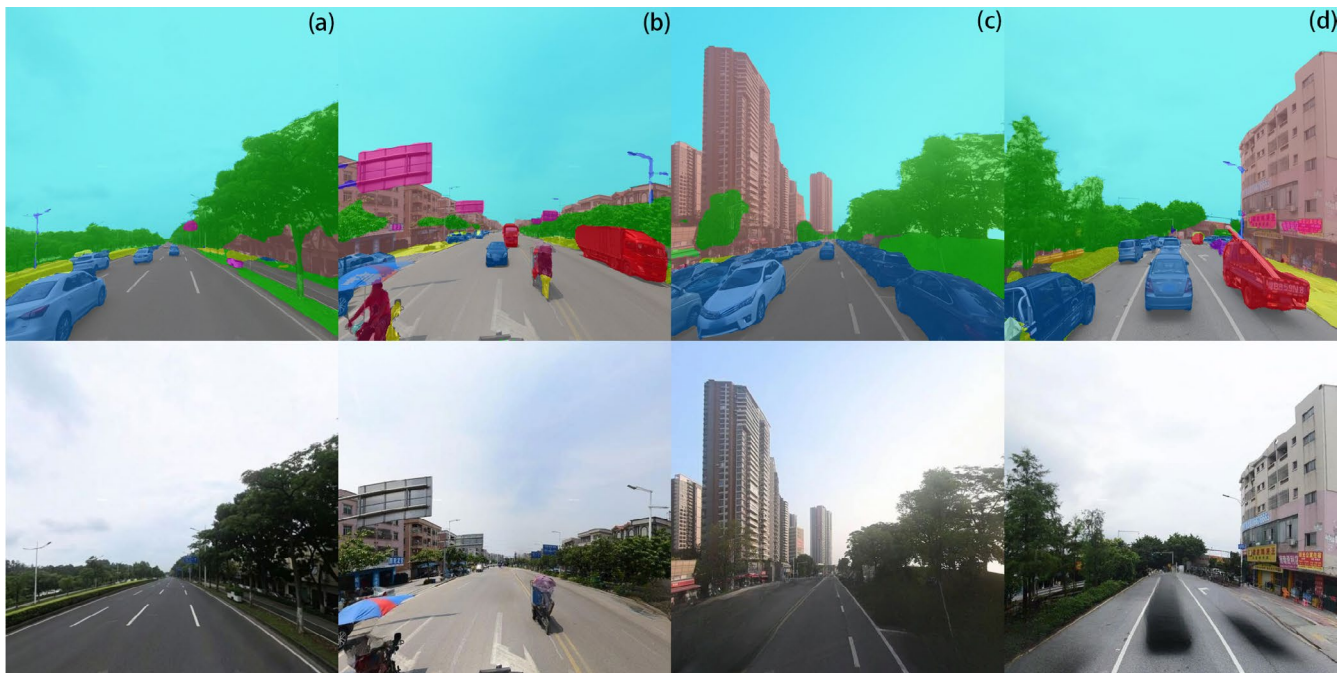


FIGURE 3 | Examples of inpainting outcomes: (a) successful removal; (b) incomplete detection (missed bicycle); (c) large blurred region (corner artifact); (d) visible vehicle shadow remaining on road.

diseases) were recruited for the experiment. Participants primarily consisted of undergraduate and graduate students, as well as faculty members in the field of Geographic Information Science, ensuring domain familiarity while maintaining diversity in age and experience. This participant composition was chosen for several reasons. First, the study required participants to complete a demanding rating task across 144 images (72 original + 72 edited) with six perceptual dimensions per image, demanding sustained attention and rating consistency over an extended period. Students and academic researchers in GIS are accustomed to systematic visual analysis and can maintain consistent judgment criteria throughout the experimental session, thereby reducing within-participant noise. Second, this sample ensures a shared baseline understanding of urban spatial concepts, which helps minimize variability arising from divergent interpretations of perceptual dimensions. Third, recruiting from a university setting enabled consistent experimental conditions and logistical feasibility. Each participant was randomly assigned two groups of images, which in fact represented the same set of urban scenes presented in two different conditions: one group contained the original street-view images with pedestrians and vehicles, and the other contained the corresponding edited images in which these dynamic elements had been removed. Each group consisted of 72 images, and every image was viewed and rated by at least eight different participants, thereby reducing individual-level noise and enhancing the robustness of the results. Each participant received a compensation of 80 RMB for completing the experiment.

3.3 | Experimental Procedure

All experiments were conducted in a well-lit laboratory room to ensure stable visual conditions. The procedure included three

phases: an introductory session, a testing session, and the formal experimental session.

3.3.1 | Introduction

Participants were welcomed and provided with an overview of the study and instructed on how to provide perceptual ratings. They were informed that the purpose of the experiment was to investigate urban perception, and that their task was to evaluate each image based on their subjective impressions across six perceptual dimensions (*wealth, safety, vibrancy, beauty, boredom, and depression*). To maintain experimental control, participants were not informed that the same scenes would appear in both original and edited versions. A brief description of the experimental process was then provided.

3.3.2 | Training

Participants are asked to complete a short trial designed to ensure that they are fully familiar with the experimental procedure. Each participant first completed a block of five practice images, which followed exactly the same procedure as the formal experiment: participants observed each street-view image for 8 s, after which the screen switched to white, and they were asked to provide ratings on the six perceptual dimensions (*wealth, safety, vibrancy, beauty, boredom, depression*). After finishing this block, participants were asked whether they felt sufficiently familiar with the procedure. If they confirmed their familiarity, they proceeded to the formal experiment; if not, they were given an additional set of five practice images, again using the same sequence of viewing and rating. All practice trials were excluded from the final analysis.

3.3.3 | Formal Experiment

The procedure is divided into participant tasks and implementation.

- **Tasks:** Each participant was asked to complete the rating of two groups of images, with each group containing 72 street-view scenes, as described in Section 3.1. One group consisted of the original images with pedestrians and vehicles, while the other group consisted of the corresponding edited images in which these dynamic elements had been removed. Participants evaluated each image on six perceptual dimensions (*wealth*, *safety*, *vibrancy*, *beauty*, *boredom*, *depression*) using a five-point Likert scale, based on their subjective impressions. To minimize memory effects, the two image groups were presented in separate sessions at least 3 days apart, following Ebbinghaus's forgetting curve (memory retention < 30% after 3 days). Within each group, the order of image presentation was fully randomized.
- **Implementation:** Before the start of each formal session on a group of street-view images, participants completed a brief training block to re-familiarize themselves with the procedure. Each trial then followed a fixed sequence: a street-view image was displayed full-screen for 8 s. This was followed by a white screen, during which participants provided their ratings. The white-screen phase served both as the rating period and as an opportunity for participants to relax. After completing their ratings, participants pressed any key to proceed to the next trial.

3.4 | Feature Extraction and Classifier Selection

3.4.1 | Scene Feature Extraction

Following established practices in computational urban perception research (e.g., Dubey et al. 2016; Zhang et al. 2018; Yang, Deng, Hu, Guan, et al. 2024; Cao et al. 2025), we thus extracted multimodal scene descriptors from three complementary perspectives—color, depth, and semantics—to characterize the structural and visual composition of each scene.

3.4.1.1 | Color Features. They describe the low-level visual tone and atmosphere of the scene. We calculated the distribution ratio of RGB components by equally dividing each color channel (R, G, and B) into 16 intervals (0–255) and computing the proportion of pixels within each range. For example, R_{red_1} represents the proportion of pixels whose red-channel values fall within the first interval (0–16), indicating the relative dominance of darker red tones in the image. In addition, we derived higher-order statistics such as color entropy and contrast, which have been shown to correlate with aesthetic and affective evaluations (Yang, Deng, Hu, Guan, et al. 2024). These RGB histogram variables are primarily included to enhance model predictive performance by capturing fine-grained chromatic distributions, rather than to serve as directly interpretable perceptual indicators. We acknowledge this trade-off and interpret them with caution in the feature importance analysis.

3.4.1.2 | Depth Features. They reflect the three-dimensional organization of urban space and viewer immersion. Depth maps

were generated using the MonoDepth2 model (a widely applied depth estimation model) (Godard et al. 2019), from which we derived mean depth ($Deep_{\text{mean}}$), standard deviation of depth ($Deep_{\text{std}}$), minimum depth ($Deep_{\text{min}}$), maximum depth ($Deep_{\text{max}}$), and median depth ($Deep_{\text{mode}}$) to quantify the spatial openness and enclosure of each scene. These indicators help explain how perceived safety or vibrancy relates to spatial structure and visibility (Zhang et al. 2018).

3.4.1.3 | Semantic Features. They capture the composition of object categories within the scene. Based on DinoV2 segmentation (Oquab et al. 2023), we calculated the proportion of key elements (buildings, trees, sidewalks, roads, sky, etc.) and summarized them into 13 semantic categories. The relative dominance of these categories (e.g., greenery ratio, sky openness) provides a high-level description of urban form and land use context (Cao et al. 2025).

3.4.2 | Classifier Selection

As outlined in Section 4.1.2, the sample sizes are imbalanced for perceptual differences on the six dimensions that appeared more frequently than others. This imbalance may lead to a bias toward the majority class in certain probability-based models. To address the issue, we chose the Synthetic Minority Over-sampling Technique (SMOTE) (<https://imbalanced-learn.org/stable/>) (Lemaître et al. 2017) to increase the number of samples in minority classes. We then conducted a pilot study to evaluate 11 commonly used classifiers for identifying perceptual variation. All classifiers were tested using default hyperparameters and an 80/20 cross-validation method. Model performance was measured using mean accuracy (acc) and $F1$ -score ($F1$), as defined in Equations (1) and (2).

$$acc = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

$$F1 = \frac{2TP}{2TP + FP + FN} \quad (2)$$

where TP , TN , FP , and FN represent true positive, true negative, false positive, and false negative, respectively.

The results of the 11 classifiers for predicting perceptual differences with multimodal features are presented in Table 2. Among all models, both the Multilayer Perceptron (MLP) and XGBoost achieved consistently strong performance across the six perceptual dimensions. The MLP model reached the highest overall scores, with accuracies of 0.879 ($F1=0.886$) for wealth, 0.877 ($F1=0.886$) for safety, and 0.887 ($F1=0.894$) for boredom. And the XGBoost model performed competitively, particularly for depression ($acc=0.892$, $F1=0.896$), boredom ($acc=0.876$, $F1=0.880$), and beauty ($acc=0.856$, $F1=0.857$), with stable results in all categories. Although the MLP slightly outperformed other models in overall metrics, XGBoost demonstrated comparable accuracy with higher computational efficiency and interpretability. Therefore, we selected XGBoost as the final model for subsequent feature importance analyses and the evaluation of the perceptual differences at the city-level.

4 | Data Analysis

4.1 | Image-Level Analysis

4.1.1 | Descriptive Statistics

According to the experimental design described in Section 3, each image (original vs. edited) was evaluated by eight independent participants. We first averaged the ratings across participants to obtain an image-level score, and then conducted descriptive analyses based on these aggregated values. Table 3 summarizes the descriptive statistics of perceptual ratings across six dimensions for original and edited street-view images. As shown in the table, we can observe that wealth, safety, boredom, and depression showed relatively stable evaluations, with mean values changing only slightly between the two conditions (wealth: 2.930 to 2.844; safety: 2.788 to 2.888; boredom: 2.858 to 2.944; depression: 2.432 to 2.428). This suggests that the removal of dynamic elements had little impact on perceptions related to wealth, safety, or negative affect. In contrast, vibrancy showed a more pronounced difference, with the average rating decreasing from 2.928 in the original images to 2.561 in the

edited ones, indicating that pedestrians and vehicles play a crucial role in conveying a sense of liveliness in urban scenes. By contrast, beauty exhibited the opposite trend: ratings increased from 2.592 to 2.738 after dynamic elements were removed, implying that the absence of pedestrians and vehicles may render scenes visually cleaner and more aesthetically pleasing.

In summary, the descriptive results suggest a differentiated effect of removing dynamic elements: vibrancy decreased while beauty increased, whereas other perceptual dimensions (wealth, safety, boredom, depression) remained relatively stable. These patterns provide initial evidence that the presence or absence of dynamic elements selectively shapes how urban scenes are perceived.

4.1.2 | Perceptual Differences at the Image Level

To further examine whether the removal of dynamic elements (vehicles and pedestrians) produced statistically significant differences in perceptual evaluations, we conducted paired comparisons

TABLE 2 | The performance of 11 classifiers for predicting perceptual differences with multimodal features.

Classifier	Wealth		Safety		Vibrancy		Beauty		Boredom		Depression	
	acc	F1	acc	F1	acc	F1	acc	F1	acc	F1	acc	F1
GBoost	0.842	0.848	0.826	0.833	0.696	0.700	0.848	0.851	0.869	0.872	0.863	0.868
LightGBM	0.825	0.834	0.810	0.812	0.700	0.694	0.843	0.846	0.858	0.863	0.853	0.857
MLP	0.879	0.886	0.877	0.886	0.708	0.718	0.862	0.869	0.887	0.894	0.874	0.886
AdaBoost	0.785	0.796	0.769	0.775	0.670	0.664	0.766	0.776	0.787	0.795	0.789	0.795
DecisionTree	0.680	0.690	0.719	0.740	0.573	0.620	0.716	0.745	0.720	0.733	0.778	0.790
LDA	0.672	0.679	0.662	0.682	0.588	0.608	0.677	0.702	0.664	0.700	0.696	0.719
KNN	0.627	0.716	0.674	0.750	0.559	0.646	0.669	0.738	0.704	0.763	0.685	0.757
LR	0.664	0.676	0.651	0.673	0.577	0.595	0.681	0.703	0.667	0.688	0.682	0.699
SVC	0.674	0.697	0.692	0.732	0.592	0.629	0.657	0.704	0.670	0.706	0.717	0.754
XGBoost	0.847	0.855	0.839	0.844	0.694	0.700	0.856	0.857	0.876	0.880	0.892	0.896
RF	0.765	0.780	0.808	0.813	0.646	0.659	0.775	0.785	0.790	0.797	0.810	0.816

Note: Bold values represent the best in each column.

TABLE 3 | Descriptive statistics of perceptual ratings for original and edited street-view images.

Dimension	Original				Edited			
	Min	Max	Ave	Std	Min	Max	Ave	Std
Wealth	1.500	4.750	2.930	0.621	1.250	4.750	2.844 (–)	0.624
Safety	1.125	4.500	2.788	0.481	1.500	4.375	2.888 (–)	0.511
Vibrancy	1.625	4.500	2.928	0.551	1.375	4.250	2.561 (↓)	0.568
Beauty	1.000	4.250	2.592	0.602	1.250	4.625	2.738 (↑)	0.653
Boredom	1.625	4.375	2.858	0.468	1.625	4.625	2.944 (–)	0.474
Depression	1.250	4.500	2.432	0.564	1.000	4.750	2.428 (–)	0.609

Note: Values with average rating changes within 0.1 are marked with a dash (–), while those exceeding 0.1 are indicated with ↑ or ↓ to represent the direction of change.

for each image based on the ratings provided by eight participants before and after editing (original vs. edited). For each perceptual dimension, we first applied the Shapiro–Wilk test, which is well-suited for small sample sizes, to assess whether the normality assumption was satisfied. When normality held, a paired-sample *t*-test was conducted; otherwise, the Wilcoxon test was used as a non-parametric alternative. The number of paired comparisons employed for each dimension is reported in Table 4. A significance threshold of $p < 0.1$ was adopted in this study due to the relatively small sample size available per image. Furthermore, employing a slightly more relaxed threshold also improves sensitivity to subtle yet potentially meaningful perceptual shifts that might otherwise be missed. Such use of trend-level criteria has precedent in perceptual and behavioral research, particularly when sample sizes are constrained (Van de Schoot and Miočević 2020). The significant results at the $p < 0.1$ level are shown in Table 4.

From the table, we can observe that almost all dimensions exhibited a certain proportion of significant differences between original and edited images, suggesting that the removal of dynamic elements did affect perception in a measurable way. The most pronounced effect was observed for vibrancy, with 30.97% of images showing significant differences and an overwhelming majority of 213 decreases versus 10 increases. This underscores the strong role of pedestrians and vehicles in conveying liveliness within urban scenes. The smallest proportion of significant results was observed for depression (8.61%), where increases and decreases were relatively balanced (25 vs. 37), suggesting heterogeneous responses across images. Between these two poles, other dimensions showed intermediate levels of sensitivity. Safety (14.44%) was most often associated with significant 82 increases versus 22 decreases, implying that the removal of dynamic elements sometimes enhanced the perceived safety of scenes. Wealth (13.61%) was primarily associated with 74 decreases versus 24 increases, suggesting that the absence of dynamic elements reduced perceived socioeconomic status. Beauty (13.47%) was strongly skewed toward 84 increases versus 13 decreases, consistent with the notion that removing pedestrians and vehicles often improves aesthetic impressions. Boredom (12.78%) tended toward 64 increases versus 28 decreases, indicating that edited scenes were sometimes experienced as more monotonous.

In summary, although the extent of significant differences varied across dimensions, the analysis demonstrates that a substantial subset of images consistently elicited perceptual changes after dynamic elements were removed. The effect was most

pronounced for vibrancy (decrease), moderately present for safety (increase), beauty (increase), wealth (decrease), and boredom (increase), and weakest and mixed for depression.

4.1.3 | Prediction of Perceptual Differences With Multimodal Features

The preceding analyses revealed consistent perceptual differences before and after removing dynamic elements; the underlying question remains why such differences occur and which visual characteristics of urban scenes drive these perceptual shifts. To explore these mechanisms, we analyzed which features contributed most to the observed perceptual differences using the XGBoost classifier (selected based on the pilot evaluation described in Section 3.4).

Using the XGBoost classifier, we further analyze the relative importance of multimodal features for predicting perceptual differences across six dimensions using the Weight metric. The Weight metric measures the number of times a feature is used for splitting across all decision trees. It reflects the model's overall reliance pattern and its decision-making behavior in distinguishing perceptual differences across street-view scenes. In the context of urban street-view perception, this metric captures how frequently different visual attributes are leveraged during the prediction process, thereby indicating the extent to which the model depends on specific types of scene features. We report the top 10 most important features in Figure 4. As shown in the figure, we can make the following observations:

For wealth, the most influential predictors were R_{light} (0.052) and R_{person} (0.051), followed by R_{blue_1} , R_{tree} , and $R_{\text{red}_{13}}$. The strong role of lighting and pedestrian-related features indicates that well-lit environments and visible human activity are essential visual cues of prosperity in urban scenes. Illuminated signage, streetlights, and people on sidewalks often imply active commercial areas or economic vibrancy. Consequently, the removal of pedestrians and vehicles weakens these contextual signals, resulting in a reduced sense of affluence. The presence of vegetation (R_{tree}) also modestly contributed to perceived wealth, suggesting that landscaped greenery often coexists with higher-quality built environments.

For safety, R_{light} again ranked highest (0.055), accompanied by R_{sidewalk} , R_{wall} , and $R_{\text{green}_{13}}$. This combination reflects the visual structure of streetscapes where sufficient illumination,

TABLE 4 | Number (*n*) and rate of images showing significant results ($p < 0.1$) across perceptual dimensions.

Dimension	Number of paired comparisons		Significant result $p < 0.1$			
	<i>t</i> -test	Wilcoxon test	<i>n</i> (↑)	<i>n</i> (↓)	<i>n</i> (Total)	Rate (%)
Wealth	217	503	24	74	98	13.61
Safety	303	417	82	22	104	14.44
Vibrancy	374	346	10	213	223	30.97
Beauty	345	375	84	13	97	13.47
Boredom	369	351	64	28	92	12.78
Depression	320	400	25	37	62	8.61

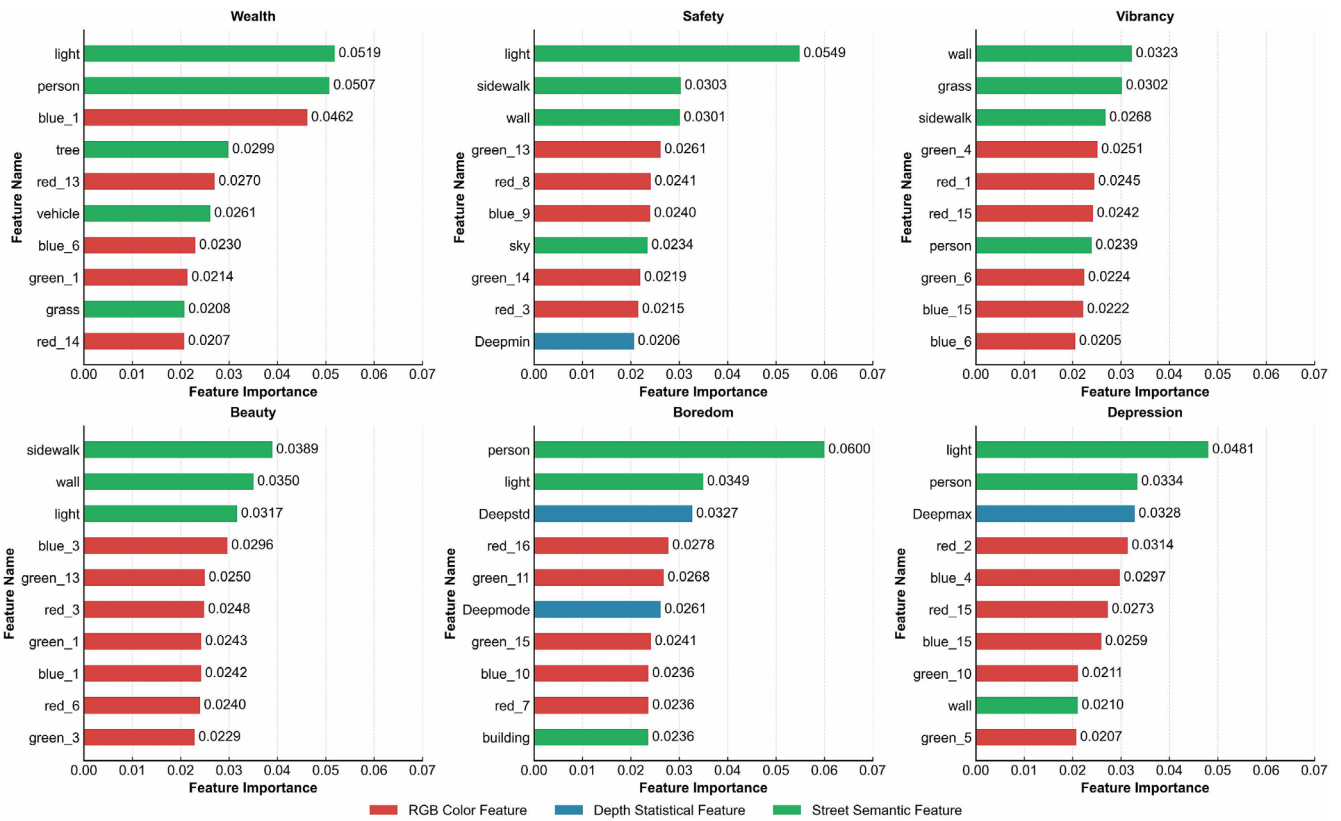


FIGURE 4 | Feature importance in XGBoost for predicting perceptual differences (edited-original) across six perceptual dimensions.

clear pedestrian pathways, and well-defined spatial boundaries contribute to feelings of security. Interestingly, $Deep_{min}$ also appeared among the top features, implying that greater spatial openness—after removing objects that previously occluded the view—may enhance perceived safety. In essence, eliminating vehicles and crowds may expose more continuous sightlines and reduce visual clutter, which participants tend to interpret as a safer, more predictable environment.

For vibrancy, R_{wall} , R_{grass} , and $R_{sidewalk}$ were the leading predictors, while R_{person} ranked prominently as well. These elements correspond to areas of social interaction and human activity, such as open sidewalks, vegetated plazas, or visually active street façades. Their importance underscores that vibrancy arises from both spatial affordances and semantic cues of movement. The disappearance of pedestrians and vehicles removes these temporal and social indicators, producing scenes that appear static and less lively. The results thus quantitatively confirm that the decline in vibrancy ratings observed in Section 4.1.2 originates from the loss of dynamic human and vehicular components that typically animate urban space.

For beauty, $R_{sidewalk}$ (0.039) and R_{wall} (0.035) were most influential, followed by R_{light} , R_{blue_3} , and R_{green_13} . These features represent organized spatial layouts and balanced chromatic compositions. The prominence of these static and structural cues indicates that, once dynamic elements are removed, the underlying spatial order becomes more visible and aesthetically pleasing. For example, removing vehicles and pedestrians often exposes architectural symmetry, coherent color gradients, and

unobstructed spatial perspectives. The increase in perceived beauty is therefore not merely a subjective bias but a direct consequence of improved visual clarity and compositional harmony within the scene.

For boredom, R_{person} (0.060) and R_{light} (0.035) ranked highest, along with depth-related indicators such as $Deep_{std}$ and $Deep_{mode}$. Scenes with fewer people and less depth variability tend to evoke monotony and visual fatigue. After removing dynamic elements, the remaining urban spaces often appear more static, lacking motion, color contrast, or social cues that sustain engagement. Reduced depth variation further limits visual exploration, creating an impression of spatial uniformity. These results explain why participants frequently rated edited scenes as more boring, as the absence of movement and human presence diminishes both semantic and spatial dynamism.

For depression, R_{light} (0.048) and R_{person} (0.033) again played dominant roles, followed by $Deep_{max}$ and low-saturation color indicators such as R_{red_2} and R_{blue_4} . The diminished contribution of light and human-related features corresponds to darker, emptier, and less inviting scenes, which may amplify feelings of desolation or confinement. In many edited images, the removal of pedestrians and vehicles left behind vacant spaces with uneven lighting and restricted depth contrast, conditions that psychologically align with isolation and emotional dullness. The role of depth features such as $Deep_{max}$ further suggests that limited spatial distance or enclosed perspectives contribute to depressive impressions by reducing the perceived openness of the urban environment.

In summary, the feature importance analysis demonstrates that the perceptual differences following the removal of dynamic elements are systematically linked to structural and semantic changes in urban imagery. Lighting and human presence consistently emerged as dominant factors shaping perceptions of wealth, vibrancy, and safety, while depth variation and color uniformity played stronger roles in feelings of boredom and depression. The absence of pedestrians and vehicles not only eliminates movement cues but also alters the visual balance of light, color, and depth, leading to measurable shifts in how people evaluate urban scenes.

4.2 | Individual-Level Analysis

4.2.1 | Descriptive Statistics

According to the experimental design described in Section 3, we collected data from 40 participants, each of whom evaluated 72 street-view scenes in both their original and edited versions.

While Section 4.1 focuses on scene-level differences to examine how individual images respond to the removal of dynamic elements, this section further investigates individual-level effects to capture consistent perceptual sensitivity across scenes. For each participant, we averaged ratings across the 72 scenes within each condition (original vs. edited) and then computed the difference between the two conditions across six perceptual dimensions. This aggregation is appropriate for our research question, as it targets individual-level perceptual sensitivity to the presence or absence of dynamic elements rather than scene-specific variability. Therefore, averaging across scenes provides a stable estimate of each participant's overall perceptual response while reducing noise induced by image-level heterogeneity.

Figure 5 illustrates the distributions of these individual-level differences. As shown in the figure, we can observe that wealth, safety, boredom, and depression showed relatively stable evaluations, with most participants exhibiting only minor differences between the two conditions. This suggests that individual judgments

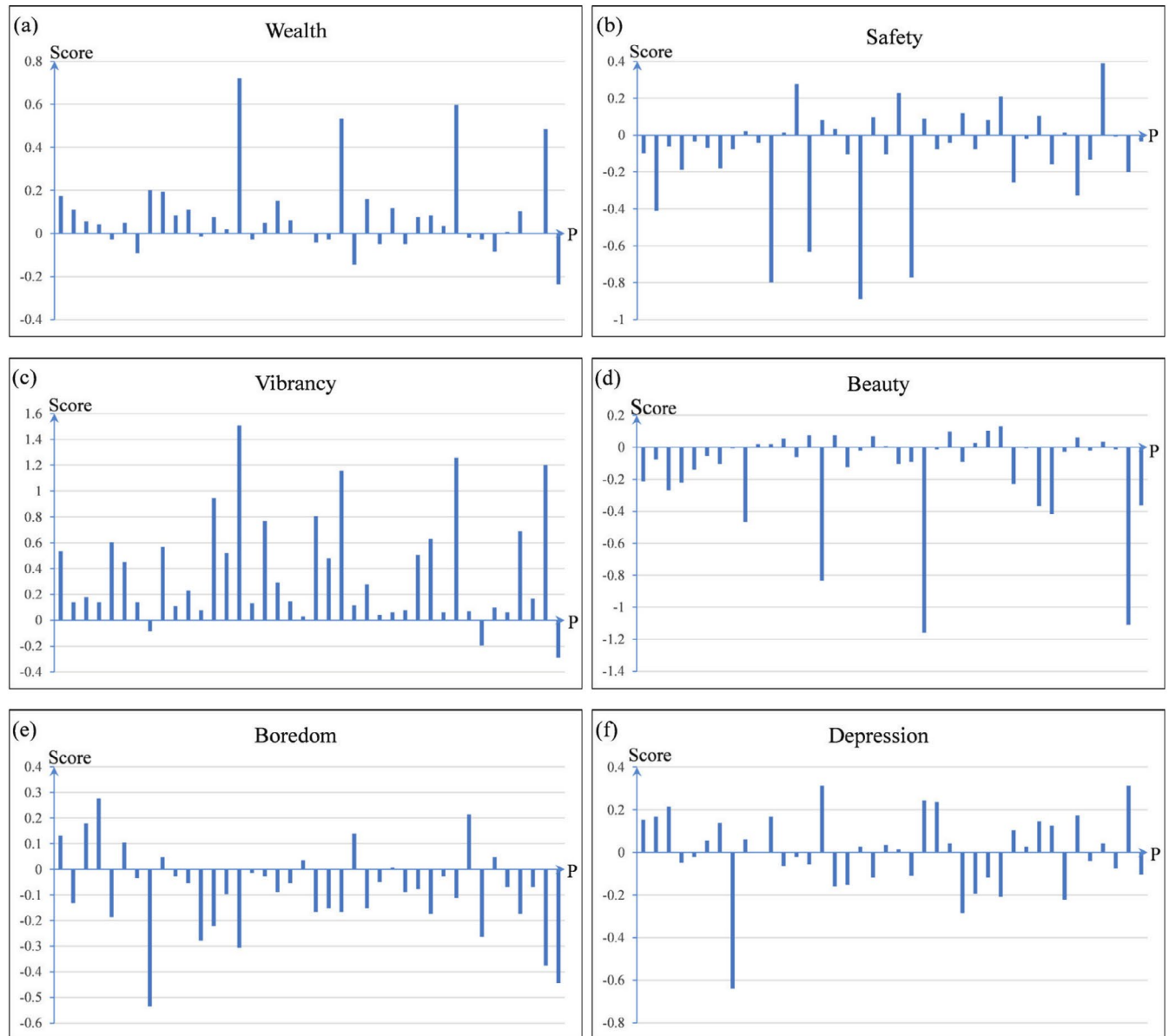


FIGURE 5 | Distributions of individual-level rating differences (original-edited) across six perceptual dimensions.

of socioeconomic status, safety, and negative affect were largely unaffected by the removal of dynamic elements. In contrast, vibrancy demonstrated a consistent decline: nearly all participants rated edited scenes lower than original ones, reinforcing the critical role of pedestrians and vehicles in conveying a sense of liveliness. Conversely, beauty showed an upward shift for a majority of participants, implying that the removal of dynamic elements often enhanced the perceived visual quality of the scenes.

In summary, these individual-level observations are highly consistent with the image-level findings: vibrancy consistently decreased while beauty increased, whereas wealth, safety, boredom, and depression remained relatively stable. This parallel indicates that the effect of removing dynamic elements is not only evident across scenes but also systematically reflected in participants' evaluations.

4.2.2 | Perceptual Differences at the Individual Level

To further examine whether the removal of dynamic elements (vehicles and pedestrians) produced statistically significant differences in perceptual evaluations at the individual level, we compared each participant's ratings for original and edited versions of the same scenes. As detailed in Section 3.2, each participant evaluated 72 scenes under each condition; we first averaged each participant's ratings across the 72 scenes within each condition to obtain a stable estimate of their general sensitivity, then computed the difference between the two conditions. The significance of these differences was assessed using the Wilcoxon signed-rank test, as the normality assumption was not satisfied (Shapiro–Wilk test). For each perceptual dimension, the distribution of paired differences (edited—original) was first examined using the Shapiro–Wilk test, which is well-suited for small sample sizes. The results indicated that the normality assumption was not satisfied; therefore, the Wilcoxon signed-rank test was employed for all dimensions. The number of participants showing significant differences in their evaluations is reported in Table 5.

From the table, we can observe that almost all dimensions exhibited a certain proportion of significant differences, indicating that the removal of dynamic elements influenced perceptual judgments for a non-negligible subset of individuals. The most pronounced effect was observed for vibrancy, with 65.0% of participants showing significant differences and the vast majority exhibiting 24 decreases versus 2 increases. This reinforces the critical role of pedestrians

and vehicles in conveying a sense of liveliness. Boredom also displayed a relatively high proportion of significant results (50.0%), with most participants reporting 17 increases versus 3 decreases, suggesting that edited scenes were often experienced as more monotonous. Safety (42.5%) showed 11 increases versus 6 decreases, implying that some participants perceived the removal of dynamic elements as enhancing the sense of safety. Depression (40.0%) was more balanced, with 7 increases versus 9 decreases, indicating heterogeneous responses across individuals. At the lower end, wealth (35.0%) was dominated by 12 decreases versus 2 increases, suggesting that socioeconomic impressions weakened when dynamic elements were removed, while beauty (35.0%) was dominated by 13 increases versus 1 decrease, implying that the absence of pedestrians and vehicles often improved aesthetic judgments.

In summary, although the extent of significant differences varied across participants, the individual-level analysis demonstrates that perceptual changes were consistently present. The strongest and most consistent effect was the decrease in vibrancy, followed by increases in boredom, safety, and beauty, while wealth tended to decrease, and depression showed mixed responses.

4.2.3 | Analysis of Moderating Factors

To further examine whether perceptual differences varied across participant characteristics, we analyzed potential moderating effects of gender. Specifically, we aimed to determine whether male and female participants differed in their sensitivity to perceptual changes caused by the removal of dynamic elements. Given that both gender and significance of perceptual difference (significant vs. non-significant per participant, as determined in Section 4.2.2) are categorical variables, the Pearson chi-square test was applied to assess whether the distribution of significant results differed between the two gender groups. This non-parametric method is suitable for identifying associations between categorical factors without assuming data normality (Sloane and Morgan 1996). The results are shown in Table 6. From the table, we can observe that only safety showed a marginally significant gender difference ($\chi^2 = 2.771$, $p = 0.096^*$), indicating that female participants were more likely to exhibit significant perceptual changes in safety evaluations after the removal of pedestrians and vehicles. This suggests that women may be more sensitive to spatial openness and social cues influencing perceived security. For other perceptual dimensions—including wealth ($p = 0.193$), vibrancy ($p = 0.285$), beauty

TABLE 5 | Number (*n*) and rate of participants showing significant differences in their perceptions ($p < 0.05$).

Dimension	<i>n</i> (↑)	<i>n</i> (↓)	<i>n</i> (Total)	Rate (%)
Wealth	2	12	14	35.0
Safety	11	6	17	42.5
Vibrancy	2	24	26	65.0
Beauty	13	1	14	35.0
Boredom	17	3	20	50.0
Depression	7	9	16	40.0

TABLE 6 | Results of the Pearson chi-square test.

Dimension	χ^2	<i>p</i>
Wealth	1.695	0.193
Safety	2.771	0.096*
Vibrancy	1.143	0.285
Beauty	1.695	0.193
Boredom	1.905	0.168
Depression	2.401	0.121

Note: * represent the 1%, 5%, and 10% significance levels, respectively.

($p=0.193$), boredom ($p=0.168$), and depression ($p=0.121$)—no significant gender differences were detected. Nevertheless, the general tendency showed that female participants had slightly higher rates of perceptual sensitivity overall.

In summary, the chi-square results suggest that gender exerts a selective moderating effect on urban perception, particularly for safety-related evaluations, while the perceptual impact of dynamic element removal remains largely consistent across genders.

5 | Evaluate the Perceptual Differences at the City Level

To examine whether the perceptual impact of dynamic elements persists at the city scale, we further conducted a city-level evaluation focusing on vibrancy, the perceptual dimension that exhibited the most substantial changes in both image-level and individual-level analyses in Section 4. Using the XGBoost model introduced in Section 4.1.3, we extended the prediction to the entire city of Dongguan. Specifically, perceptual differences were predicted for 47,963 street-view locations and 191,852 images, with four directional views (front, back, left, and right) available for each location. It should be noted that these predictions reflect the perceptual tendencies of our specific participant sample and should be interpreted as exploratory projections rather than direct measures of population-level perception.

The results indicate that removing dynamic elements leads to widespread changes in perceived vibrancy at the city scale. Among all images, 61,642 images exhibited a predicted change in vibrancy after dynamic elements were removed, accounting for 32.1% of the total street-view images. When analyzed at the location level, 35,358 locations have images that appear

to be perceived differently in terms of vibrancy, accounting for 73.7%. In particular, 16,752 locations showed perceptual changes in one view, 12,256 locations in two views, 5,422 locations in three views, and 1,028 locations in all four views, corresponding to 34.9%, 25.6%, 11.3%, and 2.1% of all locations, respectively. This distribution suggests that perceptual changes are not isolated to a small subset of scenes, but occur consistently across multiple perspectives for a substantial number of urban locations.

We also then visualized the distribution of vibrancy differences across Dongguan, as shown in Figure 6. The study area was divided into regular spatial grids. Each grid cell contains approximately 18.2 street-view locations on average. For each grid, we counted the number of images within the cell that exhibited a significant predicted change in vibrancy after dynamic elements were removed, and summarized this count as a grid-level indicator. As shown in the left panel, grids with higher counts of perceptual change are unevenly distributed across the city, forming clear spatial clusters rather than random noise. Areas with intensive road networks and active urban functions tend to exhibit higher concentrations of vibrancy change, whereas regions with more homogeneous or less active streetscapes show relatively stable perceptual outcomes. This pattern indicates that the perceptual impact of dynamic elements is spatially structured and closely related to urban activity intensity. While in the right panel, we provide four qualitative illustrations of these results through typical panoramic street-view examples. The top two cases (#40410 and #16923) show scenes whose predicted vibrancy values remain unchanged after dynamic elements are removed (Value=0), reflecting environments where static physical structures dominate perceptual impressions. In contrast, the bottom two cases (#28377 and #19144) exhibit substantial changes in vibrancy (Value=93 and Value=94, respectively). In these scenes, pedestrians and vehicles constitute

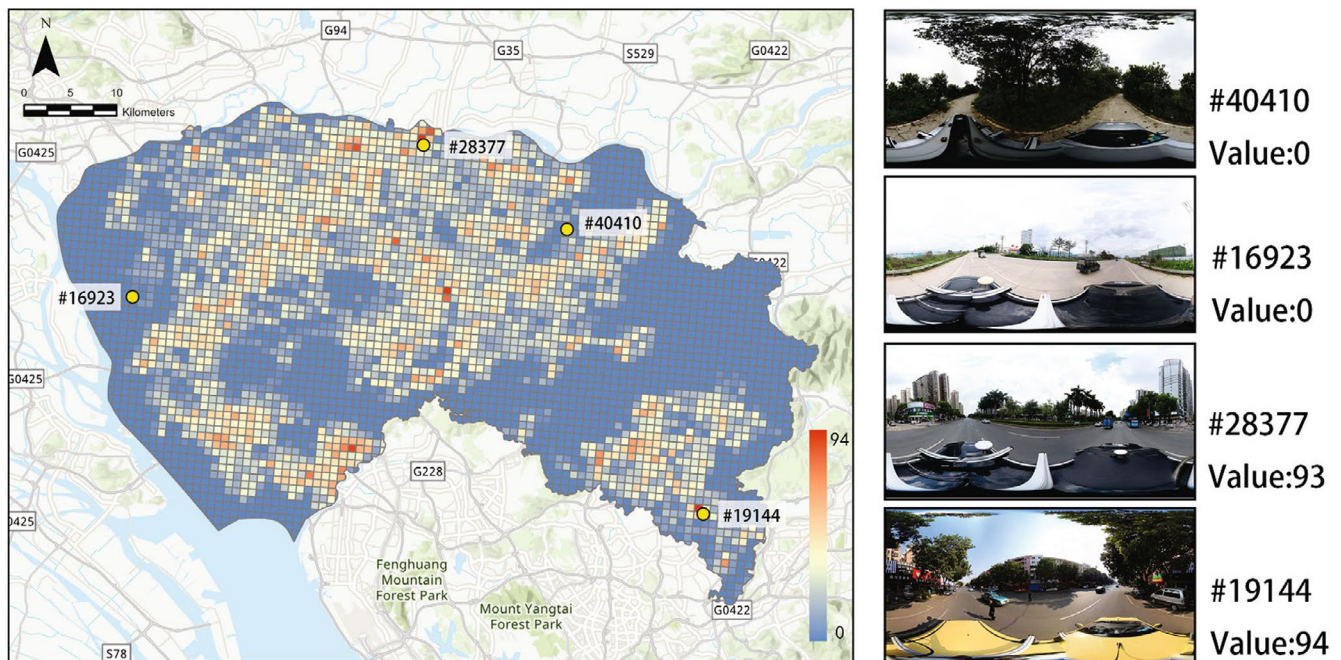


FIGURE 6 | City-level spatial distribution of predicted vibrancy changes after removing dynamic elements and representative street-view examples.

salient visual cues of social activity; once removed, the scenes become perceptually quieter and less lively despite identical underlying physical layouts. These results demonstrate that perceptual changes induced by dynamic element removal are both spatially systematic and visually interpretable. These results further support the conclusion that city-scale evaluations of urban vibrancy that ignore dynamic elements may fail to capture substantial and spatially heterogeneous aspects of human urban perception.

6 | Discussion

6.1 | The Role of Dynamic Elements in Urban Perception

Our findings highlight the differentiated impact of dynamic elements, such as pedestrians and vehicles, on urban perception. Among the six perceptual dimensions, vibrancy was most strongly influenced, with 30.97% of images and 65.0% of participants showing significant decreases after dynamic elements were removed. Vibrancy in our study reflects perceived urban vitality and intensity of human activity in a scene; thus, removing pedestrians and vehicles directly reduces the primary visual cues of movement and social presence, leading to a systematic decline in perceived vibrancy. This underscores the crucial role of human presence and mobility in conveying liveliness within urban scenes. In contrast, beauty tended to increase (13.47% of images and 35.0% of participants, the vast majority in the upward direction). Beauty here captures aesthetic preference and visual harmony of the scene; therefore, the removal of dynamic elements reduces visual complexity and foreground noise, allowing structural and compositional features (e.g., building facades, spatial layout) to become more salient, which leads to higher beauty evaluations.

For other dimensions, the patterns were more modest but still noteworthy: safety showed increases in a subset of cases (14.44% of images, mostly increases; 42.5% of participants, with 11 increases vs. 6 decreases), wealth tended to decrease (13.61% of images, mostly decreases; 35.0% of participants, 12 decreases vs. 2 increases), boredom showed limited but noticeable increases (12.78% of images and 50.0% of participants, mostly increases), and depression displayed the smallest and most mixed effects (8.61% of images; 40.0% of participants, with nearly balanced increases and decreases). The differentiated patterns can be interpreted in light of the distinct roles that dynamic elements play in shaping perceptual impressions. Wealth in this study reflects perceived socioeconomic status inferred from urban visual cues; the presence of dynamic elements may implicitly signal economic activity and urban prosperity, so their removal weakens such cues, leading to lower wealth perception. Safety reflects perceived environmental order and risk; reduced crowding and traffic may sometimes be interpreted as lower risk or calmer environments, which explains the mixed but slightly increasing tendency. Boredom and depression correspond to negative affective states associated with low stimulation and reduced engagement; thus, removing dynamic elements reduces environmental stimulation, increasing monotony and negative affect in many cases.

Collectively, these findings demonstrate that dynamic elements selectively shape how urban spaces are perceived, amplifying impressions of liveliness, often diminishing perceptions of socioeconomic status, while at times enhancing aesthetic evaluations and altering judgments of safety and boredom. Pedestrians and vehicles serve as salient markers of social activity and urban vitality, which explains why their absence significantly diminishes perceptions of vibrancy. At the same time, the removal of such elements often produces a visually cleaner environment, which may enhance aesthetic appeal and thus elevate ratings of beauty. Conversely, wealth and safety appear to depend more on static environmental cues such as building quality, greenery, or streetscape design, but the observed tendency for wealth to decrease and safety to increase indicates that dynamic elements still provide subtle cues to socioeconomic and security evaluations. These results extend previous street-view perception studies, which have predominantly focused on static physical attributes, by showing that dynamic elements constitute an additional layer of perceptual information that should not be neglected.

6.2 | Consistency Between Individual- and Image-Level Results

A second important finding is the high degree of consistency observed between image-level and individual-level analyses. At both levels, vibrancy reliably decreased following the removal of dynamic elements, whereas beauty generally increased. Other dimensions showed similar moderate patterns across levels: wealth decreased, safety and boredom increased, and depression remained mixed with no clear direction. This convergence indicates that the observed patterns are not merely artifacts of aggregation across images, but are also systematically reflected in the judgments of individual participants. Such alignment strengthens the robustness of our conclusions and highlights the pervasive influence of dynamic elements across different units of analysis.

At the same time, our analyses also reveal meaningful individual differences, particularly for safety and depression, where a considerable share of participants showed significant changes (42.5% and 40.0% respectively) even though image-level effects were modest. While aggregate results suggest relative stability, these findings imply that personal experiences, attentional preferences, or cultural contexts may moderate sensitivity to dynamic elements. This points to a promising avenue for future research: expanding the participant pool to more diverse populations and cross-cultural contexts could help disentangle the conditions under which individual variability emerges. By integrating both the consistency of group-level patterns and the heterogeneity of individual responses, our study underscores the complex but systematic ways in which dynamic elements shape urban perception.

6.3 | Implications for Urban Design and Involved Research

Our study clearly demonstrates that urban perception differs depending on whether dynamic elements such as pedestrians and

vehicles are present in street-view scenes. This evidence calls for a reconsideration of how current applications and research incorporate perceptual dimensions, as existing research has been conducted almost exclusively on static street views without accounting for these perceptual differences. In particular, we will discuss the implications from two perspectives: urban planning and design, and the related research on data-driven urban analysis.

6.3.1 | Urban Planning Implications

Our findings demonstrate that perceptions of vibrancy decrease when dynamic elements are removed, while perceptions of beauty often increase. These differentiated effects may provide actionable insights for urban design and planning. For example, if the objective is to foster a lively and vibrant atmosphere, cities may consider strengthening the visibility of dynamic elements by encouraging street activities, promoting pedestrian-friendly environments, and supporting spaces for social interaction. Conversely, if the goal is to highlight aesthetic order or cleanliness, measures such as traffic management, reducing irregular vehicle parking, or limiting overcrowded pedestrian zones could help minimize visual clutter and enhance perceived beauty. In this way, dynamic elements can be deliberately leveraged or mitigated to align with the intended perceptual goals of urban spaces.

6.3.2 | Data-Driven Urban Analysis

Our findings demonstrate that perceptual evaluations vary systematically depending on the presence or absence of dynamic elements, which implies that these factors cannot be overlooked in data-driven approaches to urban analysis. Traditional urban perception studies and machine learning models have relied almost exclusively on static street-view imagery, thereby neglecting the perceptual influence of pedestrians and vehicles. If the goal is to capture the full spectrum of human perception, dynamic elements should be explicitly considered. In some cases, they could be systematically included or excluded to test their perceptual effects in smart city applications, virtual reality environments, or simulation platforms. This process can be operationalized through the trained model described in Section 4.1.3, with the potential for greater reliability if more perceptual data are incorporated. In other cases, especially when the research focus is on the structural or environmental features of the built environment, removing dynamic elements may help achieve more consistent evaluations by reducing noise from transient social activities. This dual strategy—selective inclusion or suppression—may support “perception-oriented” urban design and opens new pathways for integrating perceptual science with digital urban analytics.

6.4 | Limitations and Future Works

Despite the contributions in this research, several limitations should be acknowledged. First, the participant sample was relatively small (40 individuals) and predominantly composed of students and faculty in Geographic Information Science, which may constrain the generalizability of the results to broader populations. Specifically, this sample is not representative of the general urban population in terms of age, educational background, occupation,

or income level. GIS professionals and students may have developed analytical habits that influence how they attend to and evaluate street-view imagery, potentially differing from how ordinary citizens perceive urban environments. While we selected this sample to ensure rating consistency and experimental control (as justified in Section 3.2), we acknowledge that the observed perceptual patterns may not fully generalize to the broader public. Therefore, the urban planning implications discussed in Section 6.3 should be interpreted as exploratory and hypothesis-generating, rather than prescriptive, and future work should replicate this experiment with more diverse and representative participant groups to validate the findings across demographic segments. Second, only the main dynamic elements in the urban scene (pedestrians and vehicles) were examined, while other potentially influential factors, such as cyclists, animals, or temporary street facilities, were not included. Third, although the inpainting process used to remove dynamic elements was carefully validated, subtle artifacts may still have influenced participant judgments. Fourth, the feature importance analysis includes low-level RGB histogram variables that are inherently limited in interpretability. While these variables were included primarily to enhance model predictive performance by capturing fine-grained chromatic distributions, their presence among the top-ranked features introduces a tension between model accuracy and mechanistic insight. We acknowledge that caution should be exercised when interpreting these particular features as indicators of perceptual mechanisms, and future work should consider replacing or complementing them with higher-level semantic color descriptors (e.g., color harmony, dominant palettes) that may offer greater interpretability without substantially sacrificing predictive power. Furthermore, as the predictive model used in the city-scale analysis (Section 5) was trained on this specific participant sample, the reported proportions (e.g., 73.7% of locations) should be interpreted with caution. The sampling bias inherent in the training data may be amplified when generalized to the entire city, and these results are best understood as indicative spatial patterns rather than precise population-level estimates.

Future research should address these limitations by incorporating more diverse participant groups, expanding the scope of dynamic elements considered, and employing cross-cultural designs. In addition, integrating computational models with perceptual experiments could further illuminate how dynamic elements interact with static urban features to shape perceptions.

7 | Conclusion

To investigate the role of dynamic elements in urban perception from street-view imagery and assess the bias introduced by treating urban scenes as static. We proposed a controlled framework that constructs paired street-view images with and without pedestrians and vehicles, enabling direct comparisons of perceptual responses. Perception experiments show that removing dynamic elements leads to a consistent decline in perceived vibrancy, while effects on other perceptual dimensions are more moderate and heterogeneous. The results further reveal notable inter-individual differences, indicating that urban perception is jointly shaped by environmental cues and individual sensitivity. By integrating multimodal visual features and machine-learning models, we demonstrate that

dynamic elements influence urban perception through both direct visual saliency and interactions with lighting and spatial structure. Extending the analysis to a city-scale dataset covering nearly 48,000 locations shows that perceptual changes induced by dynamic element removal are widespread and spatially structured. These findings suggest that perception-based urban assessments relying solely on static imagery may systematically underestimate urban liveliness. At the individual level, participants varied in their sensitivity to the presence of dynamic elements, indicating that urban perception is shaped not only by visual content but also by individual perceptual characteristics. Overall, this work highlights the necessity of incorporating dynamic elements into perception-driven urban analysis and provides a scalable methodological pathway for moving beyond static representations of cities.

Despite these findings, this study has several limitations. The perception experiment was conducted in a single city with a relatively modest number of participants, which may limit the generalizability of individual-level responses. In addition, the analysis focused on pedestrians and vehicles as dynamic elements, while other transient urban factors were not considered. Future work will extend this framework to multiple cities and cultural contexts, incorporate a broader range of dynamic urban features, and further explore temporal variations in urban perception.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The datasets generated and analyzed during the current study are not publicly available at this stage. This is due to ongoing analyses and planned extensions of the dataset. However, the data are available from the corresponding author upon reasonable request.

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